Supporting Information

In vivo antibacterial efficacy of antimicrobial peptides modified metallic implants – systematic review and meta-analysis

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Table S1

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Table S1: SYCRYL Risk of Bias tool. Each study was given "Y", "N" or "U" to represent Yes, No or Unclear respectively. Yes, represents low bias, No, represents high bias and unclear shows that there is medium bias in the care of animals during studies.

Study	Was the allocation sequence adequately generated and applied?	Were the groups similar at baseline or were they adjusted for confounders in the analysis?	Was the allocation adequately concealed?	Were the animals randomly housed during the experiment?	Were the caregivers and/or investigators blinded from knowledge which intervention each animal received during the experiment?	Were animals selected at random for outcome assessment?	Was the outcome assessor blinded?	Were incomplete outcome data adequately addressed?	Are reports of the study free of selective outcome reporting?	Was the study apparently free of other problems that could result in high risk of bias?
Adams et al., 2009	U	Y	U	U	U	U	U	U	Y	Y
Li et al., 2009	U	Y	U	U	U	U	U	U	U	Y
Gao et al., 2011	U	Y	U	U	U	U	U	U	U	Y
Sinclair et al., 2013	U	Y	U	U	U	U	U	Y	Y	Y
Windolf et al., 2014	U	Y	U	U	U	U	U	U	Y	Y
Jennings et al., 2015	U	Y	U	U	U	U	U	U	U	Y
Chen et al., 2016	U	Y	U	Y	U	U	U	Y	Y	Y
de Breij et al., 2016	U	Y	U	Y	U	U	U	Y	Y	Y
Kucharíková et al., 2016	U	Y	U	Y	U	U	U	U	U	Y
Nie et al., 2017	U	Y	U	Y	U	U	U	U	U	Y
Zhan et al., 2018	Y	Y	U	Y	U	U	U	U	U	Y
Zhang et al., 2018	U	Y	U	Y	U	Y	U	Y	Y	Y
Chen et al., 2019	U	Y	U	U	U	U	U	U	U	Y
Gao et al., 2019	U	Y	U	Y	U	U	U	Y	Y	Y
Stavrakis et al., 2019	U	Y	U	Y	Y	U	Y	Y	Y	Y
Yang et al., 2019	U	Y	U	U	U	U	U	U	U	Y
Zhang et al., 2019	U	Y	U	U	U	U	U	U	Y	Y
Chen et al., 2020	U	Y	U	U	U	U	U	U	U	Y
Xu et al., 2020	U	Y	U	Y	U	U	U	U	U	Y
Chen et al., 2021 [47]	U	Y	U	U	U	U	U	U	U	Y
Fang et al., 2021	U	Y	U	U	U	U	U	U	U	Y
Hwang et al., 2021	U	Y	U	Y	U	U	U	U	U	Y
Yang et al., 2021	Y	Y	U	Y	U	Y	U	U	U	Y
Ye et al., 2021	U	Y	U	U	U	U	U	U	U	Y

Figure S1: Forest plot of subgroup meta-analysis for assessment of AMP-coated implants bacterial infection prevention as CFU counts expression, considering AMP subgroups. Mean, standard deviation & sample size from 18 studies comparing subgroup effect size (forest plot) in control and AMP-coated groups using Hedge's g as the meta-analysis, 95% Confidence Intervals and overall effect size as a diamond. Heterogeneity, Tau^2, Chi^2 and I^2 also calculated (P<0.00001). IV, inverse variance. All bacteria used were *S. aureus* unless indicated otherwise (*E. coli* or *P. aeruginosa* (PA)).

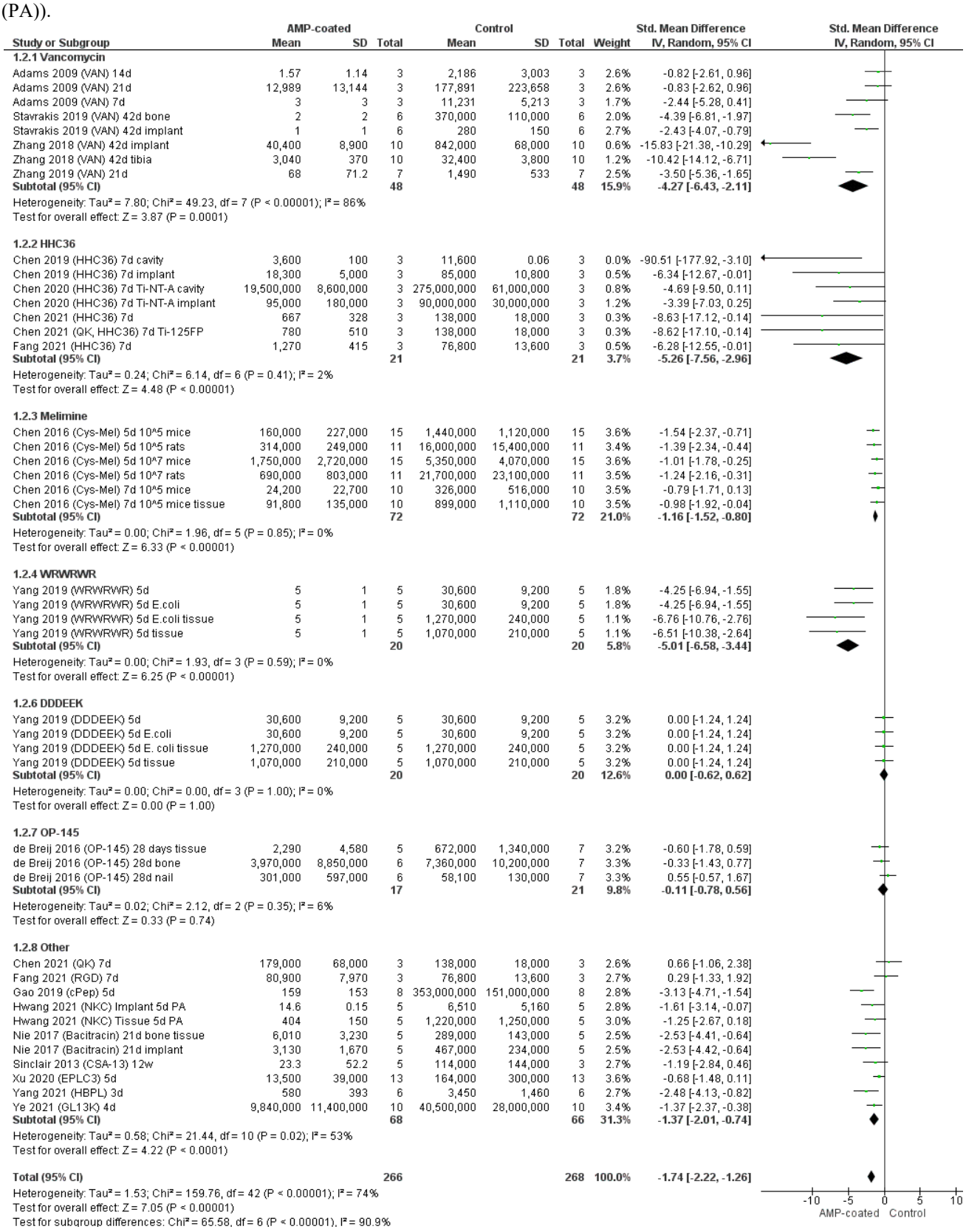

Figure S2: Forest plot of subgroup meta-analysis for assessment of AMP-coated implants on bacterial infection prevention as CFU counts expression, considering animal subgroups. Mean, standard deviation and sample size taken from 18 studies comparing effect size using Hedge's g as the meta-analysis. Effect size shown comparing control and AMP-coated groups with 95% Confidence Intervals and overall effect size as a diamond. Heterogeneity between studies, Tau^2, Chi^2 and I^2 were also calculated ($P<0.00001$). Subgroup analysis further compares overall effects sizes for different animal models used. IV, inverse variance. All bacteria used were *S. aureus* unless indicated otherwise (*E. coli* or *P. aeruginosa* (PA)).

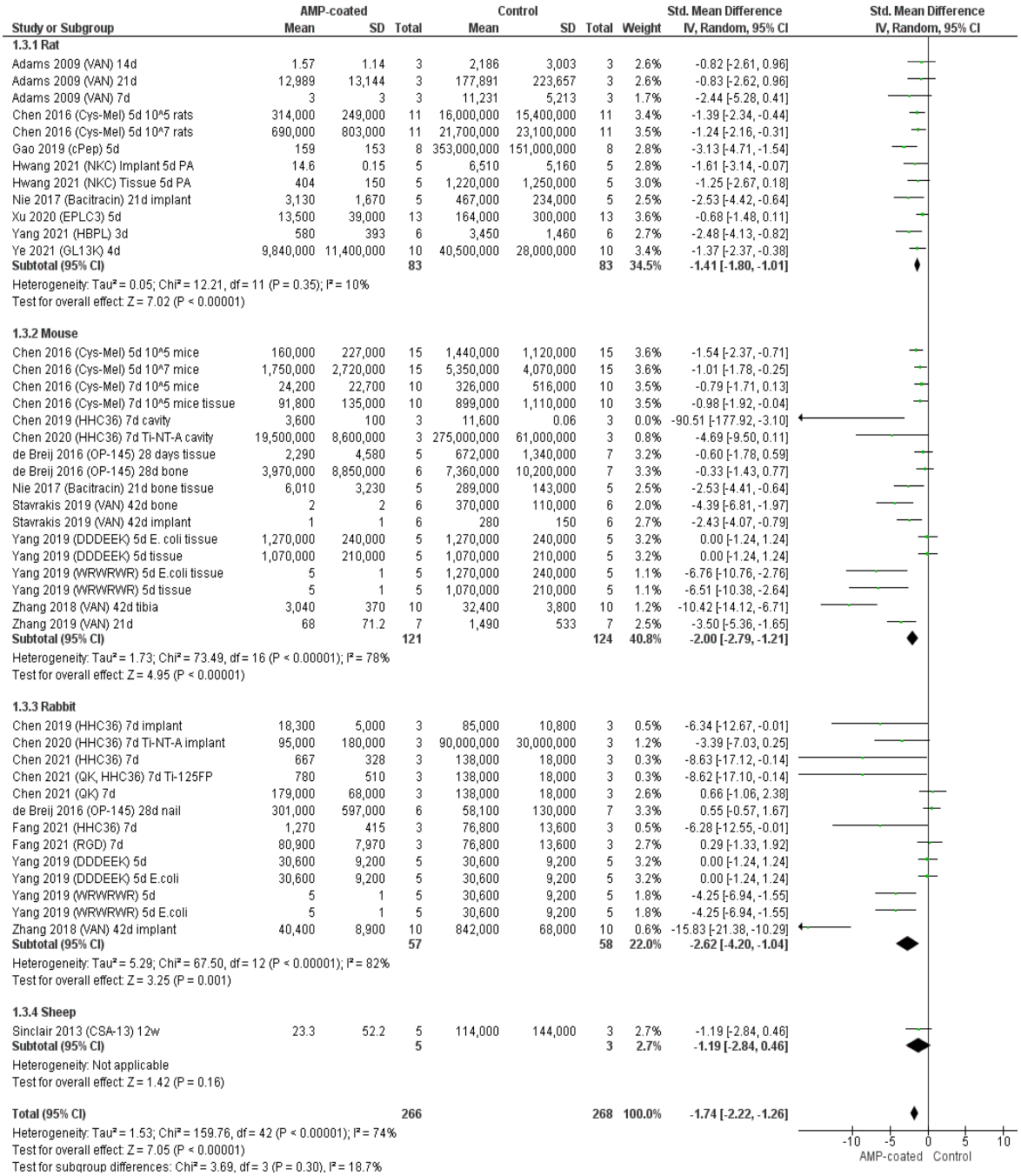

Figure S3: Forest plot of subgroup metal-analysis for assessment of AMP-coated implants on bacterial infection prevention as CFU counts expression, considering implant duration. Mean, SD and sample size taken from 18 studies to compare effect size using Hedge's g as the meta-analysis. A forest plot plotted to show effect size comparing control and AMP-coated groups with 95% Confidence Intervals and overall effect size as a diamond. Heterogeneity between studies, Tau^2, Chi^2 and I^2 were also calculated (P<0.00001). Subgroup analysis further compares overall effects sizes for implant duration in vivo. IV, inverse variance. All bacteria used were S *S. aureus* unless indicated otherwise (*E. coli* or *P. aeruginosa* (PA)).

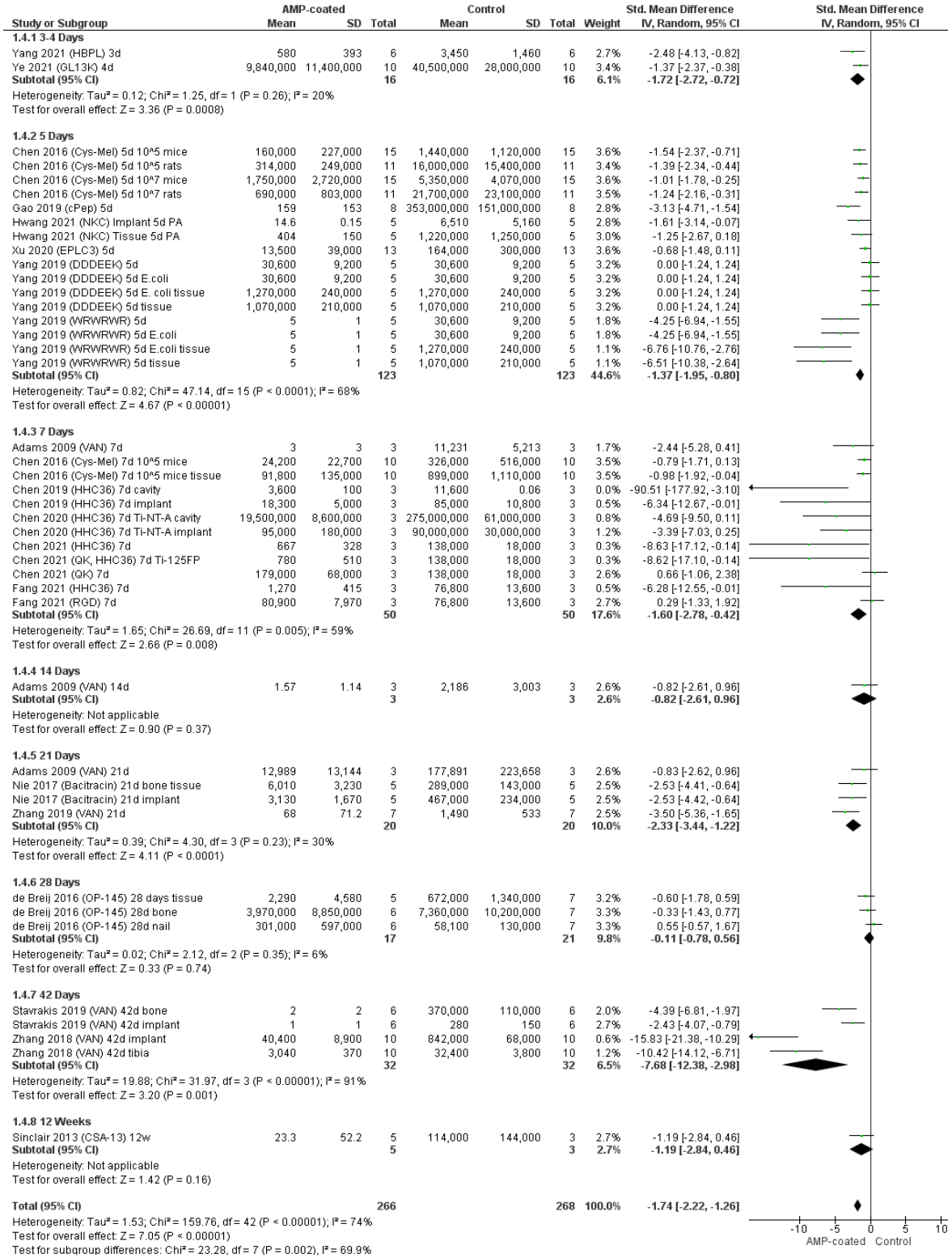
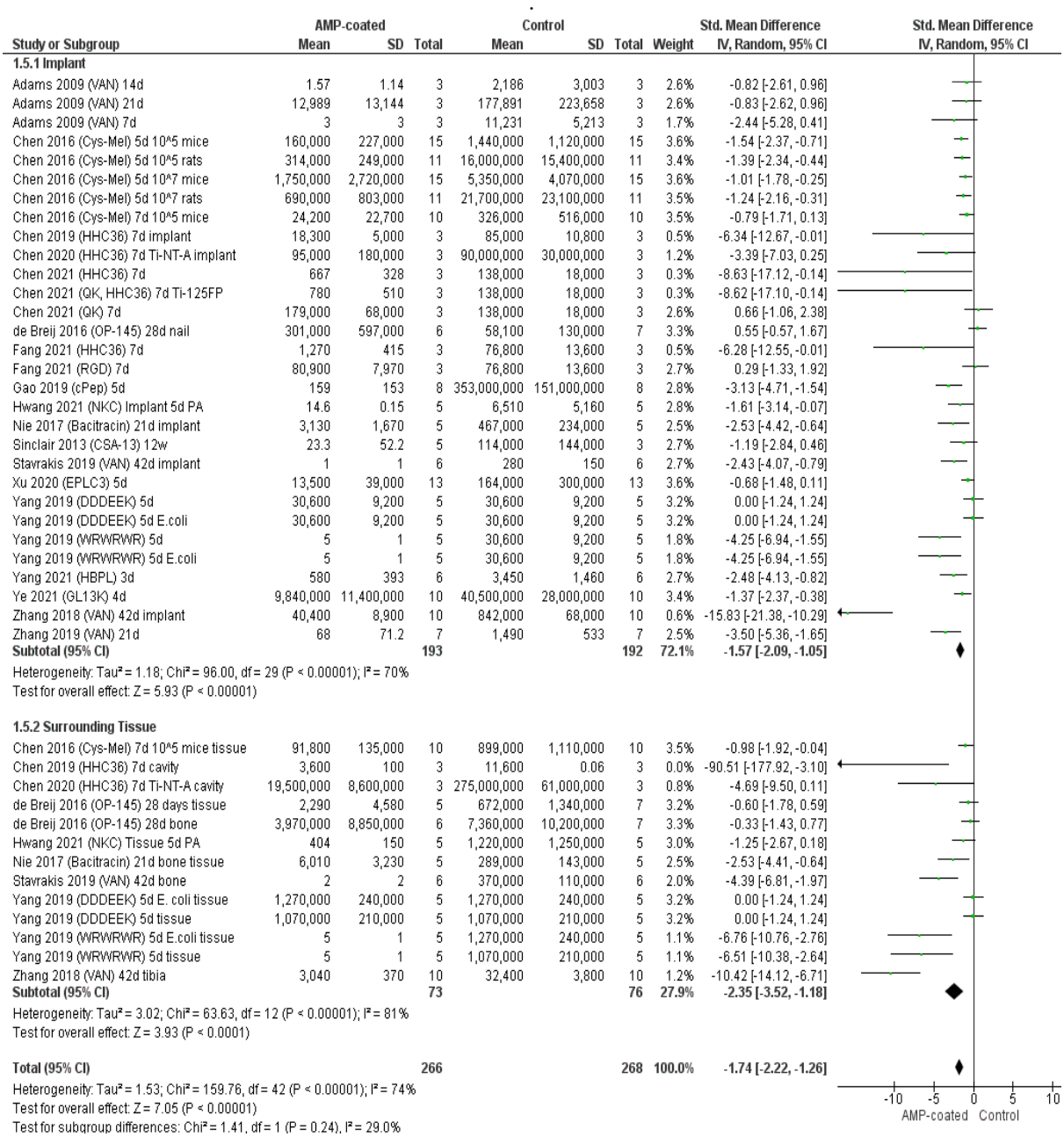
Figure S4: Forest plot of subgroup meta-anlysis for assessment of AMP-coated implants on bacterial infection prevention as CFU counts expression, considering sample type. Mean, standard deviation and sample size taken from 18 studies comparing effect size using Hedge's g as the meta-analysis. Effect size shown comparing control and AMP-coated groups with 95% Confidence Intervals and overall effect size as a diamond. Heterogeneity between studies, Tau^2, Chi^2 and I^2 were also calculated (P<0.00001). Subgroup analysis further compares overall effects sizes for different culture samples taken. IV, inverse variance. All bacteria used were *S. aureus* unless indicated otherwise (*E. coli* or *P. aeruginosa* (PA))